HW 7 Math 638 Differential Geometry II, Winter 2026

Due on Wednesday, February 25, 2026 on class.

Q7.1. ([Lee, 2ed, 12-4, p.325]) Let $V_1, \cdots, V_k$ and W be finite-dimensional real vector spaces. Prove that there is a canonical (basis-independent) isomorphism

$$V_1^* \otimes \cdots \otimes V_k^* \otimes W \simeq L(V_1, \cdots, V_k; W).$$

(In particular, this means that $V^* \times W$ is canonically isomorphic to the space $L(V; W)$ of linear maps from V to W.)

Q7.2. ([Lee, 2nd, 14-1, p. 374]) Show that covectors $\omega_1, \cdots, \omega_k$ on a finite-dimensional vector space are linearly dependent if and only if $\omega_1 \wedge \cdots \wedge \omega_k = 0$.

Q7.3. ([Lee, 2nd, 14-4(b), p. 375]) Show that two linearly independent ordered k-tuples $(v_1, \cdots, v_k)$ and $(w_1, \cdots, w_k)$ have the same span if and only if $v_1 \wedge \cdots \wedge v_k = cw_1 \wedge \cdots \wedge w_k$ for some nonzero real number c.

Q7.4. ([Lee, 2nd, 14-5, p. 375]) (Cartan's lemma) Let M^n be a smooth manifold, and let $\omega_1, \cdots, \omega_k$ be an ordered linear independent k-tuple in T_p^*M. Given covectors $\alpha^1, \cdots, \alpha_k$ in T_p^*M such that $\sum_{i=1}^k \alpha^i \wedge \omega^i = 0$, show that each α^i can be written as a linear combination of $\omega_1, \cdots, \omega_k$.

Q7.5. Compute the wedge product $\omega \wedge \eta$ where

$$\begin{aligned}\omega =& 3dx^1 \wedge dx^2 - 2dx^2 \wedge dx^3, \\ \eta =& dx^1 \wedge dx^3 \wedge dx^4 + 5dx^3 \wedge dx^4 \wedge dx^7.\end{aligned}$$

Q7.6. Let vector $v = 3\frac{\partial}{\partial x^1} - 5\frac{\partial}{\partial x^3}$ and let

$$\omega = 3dx^1 \wedge dx^4 - 2dx^2 \wedge dx^3 + 11dx^4 \wedge dx^7.$$

Compute the inner product $i_v\omega$.